

Tutorial 5.

2. a) $A = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$ $A^{-1} = \begin{pmatrix} -2 & -1 \\ 0 & 2 \end{pmatrix} \frac{1}{-4 - 0}$

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & -\frac{1}{2} \end{pmatrix}.$$

b) $A = \begin{pmatrix} -1 & 3 \\ 2 & -6 \end{pmatrix}$ $ad - bc = 6 - 6 = 0$
no inverse.

c) $A = \begin{pmatrix} 6 & 0 \\ 2 & 1 \end{pmatrix}$ $ad - bc = 6 - 0 = 6$

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 6 \end{pmatrix} \frac{1}{6} = \begin{pmatrix} \frac{1}{6} & 0 \\ -\frac{1}{3} & 1 \end{pmatrix}$$

3. a) $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ $(A - \lambda I_2) \underline{x} = \underline{0}$

Find λ : $\det(A - \lambda I_2) = 0$

$$(2 - \lambda)(2 - \lambda) - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 1 \text{ or } \lambda = 3.$$

If $\lambda = 3$ Find $\underline{x}$ st

$$A \underline{x} = 3 \underline{x}$$

$$x_1 = x_2$$

$$2x_1 + x_2 = 3x_1$$

$$x_1 + 2x_2 = 3x_2$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ will do.}$$

If $\lambda = 1$
Find $\underline{x}$ st

$$(A - I_2) \underline{x} = \underline{0}$$

$$2x_1 + x_2 = x_1$$

$$x_1 + 2x_2 = x_2$$

$$x_1 = -x_2$$

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ will do.}$$

$$3b) \det \left[\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right] = 0$$

$$\det \begin{pmatrix} -\lambda & 1 \\ -2 & -3-\lambda \end{pmatrix} = 0$$

$$\Rightarrow (-\lambda)(-3-\lambda) + 2 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$(\lambda + 2)(\lambda + 1) = 0$$

$$\lambda = -2 \quad \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2x_1 \\ -2x_1 - 3x_2 \end{pmatrix}$$

$$x_2 = -2x_1$$

$$-2x_1 - 3x_2 = -2x_2$$

$$\begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ will do.}$$

$$\lambda = -1$$

$$\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix}$$

$$x_2 = -x_1$$

$$-2x_1 - 3x_2 = -x_2$$

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ will do.}$$

$$c) A = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} \quad \det(A - \lambda I_2) = 0 \Rightarrow (2-\lambda)(-1-\lambda) - 4 = 0$$

$$-2 + \lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - \lambda - 6 = 0$$

$$(\lambda - 3)(\lambda + 2) = 0$$

$$\lambda = 3 \text{ or } \lambda = -2$$

If $\lambda = 3$ then

$$2x_1 - 4x_2 = 3x_1$$

$$-x_1 - x_2 = 3x_2$$

$$-4x_2 = x_1 \quad \begin{pmatrix} -4 \\ 1 \end{pmatrix} \text{ will do}$$

and if $\lambda = -2$

$$2x_1 - 4x_2 = -2x_1$$

$$-x_1 - x_2 = -2x_2$$

$$-4x_2 = -4x_1$$

$$x_1 = x_2$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ will do.}$$

Tutorial 5

$$\begin{aligned} 4. a) \quad y_1' &= 3y_1 + 3y_2 \\ y_2' &= 5y_1 + y_2 \end{aligned} = \begin{pmatrix} 3 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \lambda \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}$$

$$\det \begin{pmatrix} 3-\lambda & 3 \\ 5 & 1-\lambda \end{pmatrix} = 0 \Rightarrow (3-\lambda)(1-\lambda) - 15 = 0$$

$$\lambda^2 - 4\lambda + 3 - 15 = 0$$

$$\lambda^2 - 4\lambda - 12 = 0$$

$$(\lambda - 6)(\lambda + 2) = 0$$

$$\lambda = 6 \text{ or } \lambda = -2$$

$$\text{If } \lambda = 6$$

$$3y_1 + 3y_2 = 6y_1 \quad ①$$

$$5y_1 + y_2 = 6y_2 \quad ②$$

$$① \quad 3y_2 = 3y_1$$

$$y_2 = y_1$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ will do.}$$

$$\text{If } \lambda = -2$$

$$3y_1 + 3y_2 = -2y_1$$

$$5y_1 + y_2 = -2y_2$$

$$5y_1 = -3y_2$$

$$\begin{pmatrix} -3 \\ 5 \end{pmatrix} \text{ will do.}$$

$$y = c_1 e^{6t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

$$4b) \quad y_1' = -3y_1 + y_2$$

$$y_2' = y_1 - 3y_2$$

$$\text{Try } \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \lambda \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\text{Then set } \det \begin{pmatrix} -3-\lambda & 1 \\ 1 & -3-\lambda \end{pmatrix} = 0$$

$$(3+\lambda)^2 - 1 = 0$$

$$\lambda^2 + 6\lambda + 8 = 0$$

$$(\lambda+2)(\lambda+4) = 0$$

$$\lambda = -2 \text{ or } \lambda = -4$$

if $\lambda = -2$ find $\frac{y}{y}$ st

$$-3y_1 + y_2 = -2y_1$$

$$y_1 - 3y_2 = -2y_2$$

$$y_1 = y_2$$

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ will do.

$$\text{if } \lambda = -4$$

$$-3y_1 + y_2 = -4y_1$$

$$y_1 - 3y_2 = -4y_2$$

$$y_1 = -y_2$$

$\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ will do.

$$y = c_1 e^{-2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-4t}$$

$$5. a) \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 9 & -2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$= \lambda \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

10th Mar 5.

$$y_1(0) = 0$$

$$y_2(0) = 3$$

$$\det \begin{pmatrix} -2-\lambda & 1 \\ 9 & -2-\lambda \end{pmatrix} = 0 \Rightarrow (2+\lambda)^2 - 9 = 0$$

$$\lambda^2 + 4\lambda - 5 = 0$$

$$(\lambda + 5)(\lambda - 1) = 0$$

$$\text{so } \lambda = -5 \text{ or } \lambda = 1$$

if $\lambda = 1$ find y st

$$-2y_1 + y_2 = y_1$$

$$9y_1 - 2y_2 = y_2$$

$$9y_1 = 3y_2$$

$$3y_1 = y_2$$

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} \text{ will do.}$$

if $\lambda = -5$ find y st

$$-2y_1 + y_2 = -5y_1$$

$$y_2 = -3y_1$$

$$\text{or } 9y_1 - 2y_2 = -5y_2$$

$$9y_1 = -3y_2$$

$$3y_1 = -y_2$$

$$\begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ will do}$$

$$y(t) = c_1 e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 e^{-5t} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$y(0) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 \\ 3c_1 - 3c_2 \end{pmatrix}$$

$$\text{so } c_1 = -c_2$$

$$3c_1 - 3c_2 = 3$$

$$c_1 - c_2 = 1$$

$$-c_2 - c_2 = 1$$

$$c_2 = -1/2 \Rightarrow c_1 = 1/2$$

$$y(t) = +\frac{1}{2} (e^t) \begin{pmatrix} 1 \\ 3 \end{pmatrix} - \frac{1}{2} e^{-5t} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$5b) \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \lambda \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\text{Solve } \det \begin{pmatrix} 1-\lambda & 3 \\ 0 & 2-\lambda \end{pmatrix} = 0.$$

$$(1-\lambda)(2-\lambda) - 0 = 0$$

$$\text{so } \lambda = 1 \text{ or } \lambda = 2.$$

$$\text{if } \lambda = 1 \quad y_1 + 3y_2 = y_1$$

$$2y_2 = y_2$$

$$\text{so } y_2 = 0, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ will do.}$$

$$\text{if } \lambda = 2.$$

$$y_1 + 3y_2 = 2y_1 \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ will do.}$$

$$2y_2 = 2y_2$$

$$3y_2 = y_1$$

$$y(t) = c_1 e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$y(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 + 3c_2 \\ 0 + c_2 \end{pmatrix} \quad \text{so } c_2 = 1 \text{ and } c_1 = -1$$

$$y(t) = e^t \begin{pmatrix} -1 \\ 0 \end{pmatrix} + e^{2t} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Tutorial 5

b. a) $y'' - 7y' + 10y = 0$

$$y'' = -10y + 7y'$$

$$\underline{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} y \\ y' \end{pmatrix}$$

$$\underline{v}' = \begin{pmatrix} y' \\ y'' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -10 & 7 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = \lambda \begin{pmatrix} y \\ y' \end{pmatrix}$$

solve λ given $\det \begin{pmatrix} -\lambda & 1 \\ -10 & 7-\lambda \end{pmatrix} = 0$

$$(\lambda - 7)(\lambda) + 10 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\lambda = 2 \text{ or } \lambda = 5$$

if $\lambda = 2$ then

$$\begin{pmatrix} y' \\ -10y + 7y' \end{pmatrix} = \begin{pmatrix} 2y \\ 2y' \end{pmatrix}$$

$$y' = 2y$$

$$-10y = -5y'$$

$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ will do

if $\lambda = 5$ then

$$\begin{pmatrix} y' \\ -10y + 7y' \end{pmatrix} = \begin{pmatrix} 5y \\ 5y' \end{pmatrix}$$

$$-10y = -2y'$$

$$5y = y'$$

$\begin{pmatrix} 1 \\ 5 \end{pmatrix}$ will do.

or $v_2 = 5v_1$

$$-10v_1 + 7v_2 = 5v_2$$

$\begin{pmatrix} 1 \\ 5 \end{pmatrix}$ will do.

or $v_2 = 2v_1$
 $-10v_1 + 7v_2 = 2v_2$

$$\underline{v}(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{5t} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} y(t) \\ y'(t) \end{pmatrix} \text{ so } y(t) = c_1 e^{2t} + c_2 e^{5t}$$

$$6b) \quad y'' + 6y' + 8y = 0.$$

$$\underline{v}(t) = \begin{pmatrix} v_1(t) \\ v_2(t) \end{pmatrix} = \begin{pmatrix} y(t) \\ y'(t) \end{pmatrix}$$

$$\underline{v}'(t) = \begin{pmatrix} v_1'(t) \\ v_2'(t) \end{pmatrix} = \begin{pmatrix} y'(t) \\ y''(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -8 & -6 \end{pmatrix} \begin{pmatrix} v_1(t) \\ v_2(t) \end{pmatrix}$$

$$\det \begin{pmatrix} -\lambda & 1 \\ -8 & -6-\lambda \end{pmatrix} = 0$$

$$(\lambda + 6)(\lambda) + 8 = 0$$

$$\lambda^2 + 6\lambda + 8 = 0$$

$$(\lambda + 2)(\lambda + 4) = 0$$

$$\text{so } \lambda = -2, \lambda = -4$$

Find $\underline{v}(t)$ so that

$$\begin{pmatrix} 0 & 1 \\ -8 & -6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = -2 \underline{v}$$

$$v_2 = -2v_1$$

$$-8v_1 - 6v_2 = -2v_2$$

$$\begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ will do.}$$

Find $\underline{v}(t)$ such that

$$\begin{pmatrix} 0 & 1 \\ -8 & -6 \end{pmatrix} \underline{v} = -4 \underline{v}$$

$$v_2 = -4v_1$$

$$-8v_1 - 6v_2 = -4v_2$$

$$\begin{pmatrix} 1 \\ -4 \end{pmatrix} \text{ will do.}$$

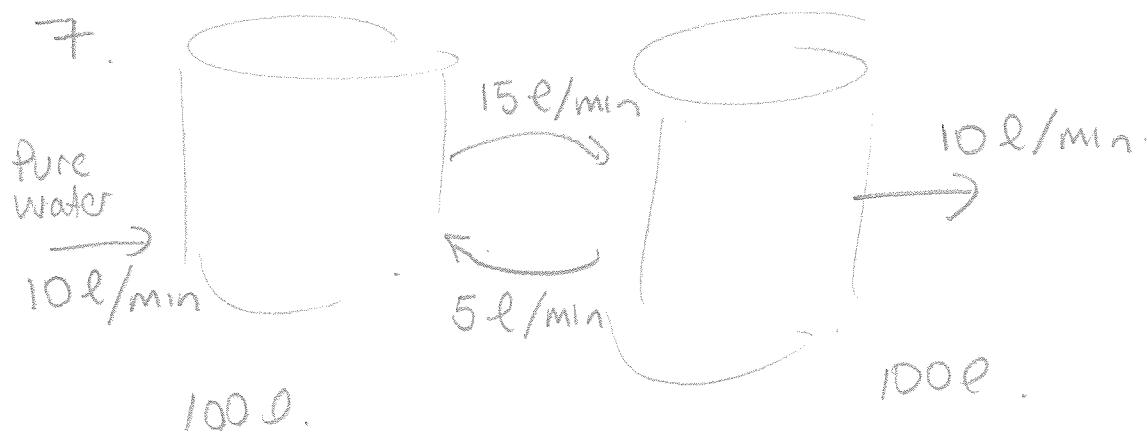
$$\underline{v}(t) = c_1 e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 e^{-4t} \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$= \begin{pmatrix} y(t) \\ y'(t) \end{pmatrix}$$

$$\text{so } y(t) = c_1 e^{-2t} + c_2 e^{-4t}$$

Tutorial 5.

7.



Tank 1

Tank 2

initially; 50 kg salt.

$y_i(t)$ is conc

of salt in Tank i , $i=1, 2$.

$$y_1'(t) = \frac{5}{100} y_2(t) - \frac{15}{100} y_1(t)$$

$$y_2'(t) = \frac{15}{100} y_1(t) - \frac{5}{100} y_2(t) - \frac{10}{100} y_2(t)$$

$$= \frac{15}{100} y_1(t) - \frac{15}{100} y_2(t)$$

$$\begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix} = \begin{pmatrix} -0.15 y_1 + 0.05 y_2 \\ 0.15 y_1 - 0.15 y_2 \end{pmatrix} = \begin{pmatrix} -0.15 & 0.05 \\ 0.15 & -0.15 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Find λ :

$$\det \begin{pmatrix} -0.15 - \lambda & 0.05 \\ 0.15 & -0.15 - \lambda \end{pmatrix} = 0$$

$$(\lambda + 0.15)^2 - 0.05(0.15) = 0$$

$$\lambda^2 + 0.3\lambda + 0.15^2 - 0.05(0.15) = 0$$

Tutorial 5.

b. c) $y'' + 5y' + 6y = 0.$

$$\underline{v}(t) = \begin{pmatrix} v_1(t) \\ v_2(t) \end{pmatrix} = \begin{pmatrix} y(t) \\ y'(t) \end{pmatrix}$$

$$\underline{v}'(t) = \begin{pmatrix} v_1'(t) \\ v_2'(t) \end{pmatrix} = \begin{pmatrix} y'(t) \\ y''(t) \end{pmatrix}$$

$$\underline{v}'(t) = \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix} \begin{pmatrix} v_1(t) \\ v_2(t) \end{pmatrix}$$

say $\det \begin{pmatrix} -\lambda & 1 \\ -6 & -5-\lambda \end{pmatrix} = 0$

if $\lambda = -3$.

$$(5+\lambda)(\lambda) + 6 = 0$$

$$\lambda^2 + 5\lambda + 6 = 0$$

$$(\lambda + 2)(\lambda + 3) = 0$$

so $\lambda = -2$ or $\lambda = -3$

$$v_2 = -3v_1 \quad \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

will do.

if $\lambda = -2$ then

$$v_2 = -2v_1$$

$$-6v_1 - 5v_2 = -2v_2$$

$$\begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ will do.}$$

$$\text{so } \underline{v}(t) = c_1 e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\text{so } y(t) = c_1 e^{-2t} + c_2 e^{-3t}$$

Total 5

7. ctd. $\lambda^2 + 0.3\lambda + (0.15)(0.15 - 0.05) = 0$

$$\lambda^2 + 0.3\lambda + (0.15)(0.1) = 0$$

$$\lambda = \frac{-0.3 \pm \sqrt{(0.3)^2 - 4(0.15)(0.1)}}{2}$$

2.

$$= \frac{-0.3 \pm \sqrt{0.03}}{2} = \frac{-0.3 \pm 0.173}{2}$$

$$\lambda = -0.237 \text{ or } -0.0634$$

so if $\lambda = -0.237$

$$-0.15y_1 + 0.05y_2 = -0.237y_1$$

$$0.05y_2 = -0.087y_1$$

$$y_2 = -1.74y_1$$

$$\begin{pmatrix} 1 \\ -1.74 \end{pmatrix}$$

if $\lambda = -0.0634$

$$-0.15y_1 + 0.05y_2 = -0.0634y_1$$

$$0.05y_2 = 0.0866y_1 \quad \left(\frac{0.173}{2} \right) \quad \begin{pmatrix} 1 \\ 1.74 \end{pmatrix}$$

$$y_2 = 1.732y_1$$

$$y(t) = c_1 e^{-0.0634t} \begin{pmatrix} 1 \\ 1.74 \end{pmatrix} + c_2 e^{-0.237t} \begin{pmatrix} 1 \\ -1.74 \end{pmatrix}$$

$$\text{at } t=0 \quad C_1 + C_2 = 0 \quad C_2 = -C_1$$

$$1.74 (C_1 - C_2) = 50$$

$$1.74 \cdot 2C_1 = 50$$

$$C_1 = \frac{25}{1.74}$$

$$1.74^2 = 3.0276$$

1.74 must be
approx for $\sqrt{3}$

$$\text{so } y_1(t) = \frac{25}{\sqrt{3}} e^{-0.0634t} - \frac{25}{\sqrt{3}} e^{-0.237t}$$

$$\lambda = \frac{-0.3 \pm \sqrt{(0.3)^2 - 4(0.15)(0.1)}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.09 - 0.06}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.03}}{2}$$

$$= \frac{-0.3 \pm \sqrt{0.01} \sqrt{3}}{2} = \frac{-0.3 \pm 0.1\sqrt{3}}{2}$$

$$= -0.15 \pm 0.05\sqrt{3}$$

Tutorial 5:

$$\lambda = -0.15 + 0.05\sqrt{3}$$

find

$$-0.15y_1 + 0.05y_2 = -0.15y_1 + 0.05\sqrt{3}y_1$$

$$y_2 = \sqrt{3}y_1 \quad \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$\lambda = -0.15 - 0.05\sqrt{3}$$

$$\text{Then } -0.15y_1 + 0.05y_2 = -0.15y_1 - 0.05\sqrt{3}y_1$$

$$\text{so } y_2 = -\sqrt{3}y_1 \quad \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}$$

$$\text{so } y(t) = c_1 e^{-0.0634t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + c_2 e^{-0.236t} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$\text{and } y(0) = \begin{pmatrix} 0 \\ 50 \end{pmatrix} \quad \text{so } c_1 + c_2 = 0$$

$$\text{and } 50 = -\sqrt{3}c_1 + \sqrt{3}(-c_1)$$

$$\text{so } \frac{50}{-2\sqrt{3}} = c_1 = -\frac{25}{\sqrt{3}}$$

$$\text{and } c_2 = \frac{25}{\sqrt{3}}$$

$$y(t) = -\frac{25}{\sqrt{3}} e^{-0.0634t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + \frac{25}{\sqrt{3}} e^{-0.236t} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$$

$$y_1(t) = -\frac{25}{\sqrt{3}} e^{-0.0634t} + \frac{25}{\sqrt{3}} e^{-0.236t}$$

